

Predicting the Results of Professional Tennis Matches with Machine Learning

SPEC 1100 Applied Machine Learning

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1 Introduction

Tennis is a popular sport where the one-on-one nature of competition provides an ideal environment for sports analytics, as match outcomes are determined by a more controlled set of variables than team-based sports. While professional tennis has benefited from rigorous record-keeping and a wealth of statistics about every aspect of the game, accurately predicting match results remains a complex challenge due to the intricate interactions between player skills, form, physical and psychological state, as well as environmental factors such as weather, court surface type, and tennis ball characteristics that can significantly influence the performance of players and the outcome of a match [1].

Professional tennis is governed by several organisations. Among them, the Association of Tennis Professionals (ATP), which manages the men's circuit, and the Women's Tennis Association (WTA), which oversees the women's tour, are the most important. These organisations maintain their official ranking systems that are used to determine tournament eligibility, entry qualification, and player seeding. The official rankings for both tours are based on a 52-week cumulative rolling points system. Under this framework, players earn points for each match won in an official tournament, with wins in later rounds valued more highly than those in earlier rounds. Points earned in a specific tournament remain on a player's record for 52 weeks before expiring. The number of points awarded for each win is strictly determined by the event level or category, as well as the round the match is played on.

These point systems have changed several times over the years. A brief summary of the current event levels on the ATP Tour and some examples of the number of points awarded to players in different rounds is shown on Table 1. For instance, the champion of one of the four annual Grand Slam events (Australian Open, Roland Garros, Wimbledon, and the US Open) receive 2,000 points, whereas the winner of a first round match that does not advance any further on these tournaments receive 50 ATP points. The ATP Finals is the season-ending event where only 8 of the top-ranked players are eligible

to play. It has a unique format not seen in any other ATP Tour events where players are separated into two groups of four and play three round-robin matches (each worth 200 points), with the top two performers from each group proceeding to play in the knock-out phase. A player can collect up to 1,500 points in the ATP Finals.

Table 1: Summary of the current event categories in the main ATP Tour and the number of points awarded to event champion, runner-up, and a player who wins the first round but does not advance any further.

Category	Events	Champion	Runner-up	First round winner
Grand Slam	4	2,000	1,300	50
ATP Finals	1	up to 1,500	up to 1,000	200
Masters 1000	9	1,000	650	up to 50
ATP 500	16	500	330	up to 50
ATP 250	30	250	165	up to 25

While the ATP and WTA rankings are a good indication of the current strengths and abilities of tennis players, and can generally be used to predict the winner of a professional tennis match, they obviously do not capture the full story and are far from being a reliable predicting measure, particularly when the difference in rankings between players is small.

One area that the official ranking systems completely ignore is the court surface. Currently, professional tennis is played on three types of courts: clay, grass, and hard courts. Each surface favours a different type of play and strategy. For instance, clay courts are characterised by slower pace and a higher ball bounce, demanding superior endurance and favouring a more defensive type of play. Grass courts are the polar opposite and favour a more attacking style. Players often have a type of court where they are likely to be more successful. For instance, Rafael Nadal is regarded as the best of all time on clay, having won a record 14 Roland Garros titles, but only won two titles on the grass of Wimbledon. In contrast, Roger Federer is the male player with the most Wimbledon titles (8), but won only one Grand Slam on clay.

The focus of this research is to develop a framework that prioritises the curation of high-quality data and the engineering of domain-driven features that can capture the multiple facets of the game and be used to predict the results of professional tennis matches. Central to this approach is the use of “enhanced covariates” — variables that are not directly observed but are derived through mathematical transformations or separate statistical models.

2 Previous Work

See [2] for the details of the modern machine learning approaches and different forecast strategies on Grand Slam tennis tournaments.

Maybe see [3] for the details of the statistical enhanced learning for modeling and prediction tennis matches at Grand Slam tournaments.

See [4] for the details of the machine learning framework for forecasting professional tennis matches using ATP statistics and betting odds.

See [5] for the details of the novel comparative study of NNAR approach with linear stochastic time series models in predicting tennis player’s performance.

See [6] for the details of the modeling and prediction of tennis matches at Grand Slam tournaments.

See [7] for the details of the forecasting tennis match results using the Bradley-Terry model.

See [8] for the details of the study of forecasting tennis matches via the Glicko model.

See [9] for the details of the Weighted Elo rating method.

See [10] for the details of the how well do Elo-based ratings predict professional tennis matches.

See [11] for the details of the sports prediction and betting models in the machine learning age: The case of tennis.

See [12] for the details of the random forest model identifies serve strength as a key predictor of tennis match outcome.

See [13] for the details of the predicting the outcome of a tennis tournament: Based on both data and judgments.

See [1] for the details of the a common-opponent stochastic model for predicting the outcome of professional tennis matches.

See [14] for the details of the A Bradley-Terry type model for forecasting tennis match results.

See [15] for the details of the tennis winner prediction based on time-series history with neural modeling.

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